

The reaction ensemble method for the computer simulation of chemical and phase equilibria. I. Theory and basic examples

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We present a new efficient Monte Carlo method for the molecular-based computer simulation of chemical systems undergoing any combination of reaction and phase equilibria. The method requires only a knowledge of the species intermolecular potentials and their ideal-gas properties, in addition to specification of the system stoichiometry and thermodynamic constraints. It avoids the calculation of chemical potentials and fugacities, as is similarly the case for the Gibbs ensemble method for phase equilibrium simulations. The method's simplicity allows it to be easily used for situations involving any number of simultaneous chemical reactions, reactions that do not conserve the total number of molecules, and reactions occurring within or between phases. The basic theory of the method is presented, its relationship to other approaches is discussed, and applications to several simple example systems are illustrated.

I. INTRODUCTION

The numerical calculation of compositions of complex chemical systems undergoing multiphase reaction equilibria is an important problem in chemistry and chemical engineering. Although the basic principles were essentially formulated long ago by Gibbs, the problem continues to attract interest, due both to its practical importance and to its interesting theoretical¹ and computational features.²⁻⁴

A statement of the general phase and reaction equilibrium problem at specified temperature T and pressure P is³

$$\min G(\mathbf{n}) = \sum_{i=1}^s n_i^\alpha \mu_i^\alpha(T, P, \mathbf{n}^\alpha) \quad (1)$$

subject to

$$\sum_{i=1}^s a_{ji} n_i^\alpha = b_j; \quad j = 1, 2, \dots, m, \quad (2)$$

$$n_i \geq 0. \quad (3)$$

G is the system Gibbs free energy, n_i^α is the molar amount of species i in phase α , whose chemical potential is μ_i^α . The latter quantity is a function of T, P , and the composition of the phase α in which the species is present. a_{ji} is the number of atoms of element j per molecule of species i , and b_j is the total amount of element j in the system. The total number of species is s and the number of elements is m . To avoid unnecessary complications, we assume that Eqs. (2) are linearly independent and that there exists more than one feasible solution.

It is important to note that we define a *species* as a substance whose molecules consist of specified numbers of atomic elements, and which exists in a specified phase. For example, $\text{H}_2\text{O}(l)$ and $\text{H}_2\text{O}(g)$ are considered to be different species. This allows the above formulation to subsume

the problem of phase equilibrium since this problem results when the elements a_{ji} comprise the identity matrix. Finally, we remark that the formulation of the equilibrium problem at specified (T, V) is identical to the above (T, P) formulation, except for the fact that the objective function is the Helmholtz free energy A , the chemical potentials must be expressed as functions of $(T, V, \mathbf{n})$, and each phase must be at the same pressure.

Numerical calculations of chemical and phase equilibria require the specification of chemical potential models for the species in each phase. In practice, most such models are semiempirical, being based typically on either empirical equations of state, or assumed simple analytical forms (for a general discussion of such models, see, e.g., Sandler⁵).

Although chemical potentials may be viewed as macroscopic consequences of the microscopic interactions between the constituent molecules of the phase at the molecular level, their calculation using a molecular-based approach has not been employed extensively due to the inherent difficulty of this statistical mechanical problem. Although recent progress has been made for pure fluids using integral equation theories,⁶ the techniques are still too numerically unwieldy and inaccurate for mixtures of any complexity. Attempts have also been made to use integral equations to model simple chemical equilibria directly using specially chosen intermolecular potential models.⁷

Computer simulation techniques provide, in principle, a powerful experimental tool for the calculation of macroscopic fluid properties in terms of their molecular properties. However, although there is much current interest in directly calculating chemical potentials using such techniques,⁸ these approaches are currently not feasible except for the simplest of chemical systems.

Three previous groups of researchers have attempted to directly study relatively simple chemical reacting systems (those involving a single chemical reaction) using molecular-based computer simulation techniques. Coker and Watts⁹ performed a Monte Carlo simulation of equi-

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librium for the liquid-phase chemical reaction



using a modification of the grand canonical ensemble in conjunction with specific models for the intermolecular potential functions. Kofke and Glandt¹⁰ later showed that this approach contained some errors, and they suggested an alternative method involving a semigrand ensemble and requiring the calculation of a single species chemical potential in the course of the simulations. Their approach was also applicable in principle to systems undergoing phase, as well as chemical reaction equilibria, although it is limited to the case of a reaction in which the total number of molecules is conserved. They applied their method to reaction (4). Finally, in recent important work, Shaw¹¹⁻¹³ has developed a simulation algorithm to study chemical equilibrium and applied it to the gas-phase reaction



Shaw's approach proceeds by implementing a Markov chain derived from the microscopic viewpoint of indistinguishability of the particles in the simulations, resulting in a set of rather complex formulas. Possible extensions were briefly discussed¹³ to the cases of multiple reactions and phases, and of reactions which do not conserve the total number of molecules (the latter requiring the artificial concept of a "null particle"). However, no calculations were presented for any of these more complex systems.

The purpose of this series of papers is to derive and apply a new and simple Monte Carlo statistical mechanical algorithm for the study of general chemically reacting systems based on a knowledge of the underlying molecular interactions, which we call the *reaction ensemble* method. The method is applicable to calculations at either fixed (T, V) or (P, T) , and to systems undergoing any number of simultaneous reactions occurring in any distribution of phases. The reaction ensemble method employs a Markov chain sampling procedure based on the viewpoint of the distinguishability of molecules in the simulations. The method exploits the direct link between the reaction stoichiometry and the elemental abundance constraints of Eq. (2), and its resulting simplicity contrasts with the more complex procedures required by the algorithm of Shaw.¹¹⁻¹³ No direct calculation of either chemical potentials or fugacities is required. The method's simplicity allows its straightforward generalization and application to many cases of chemical interest not considered previously, including situations when the reactions do not conserve the total number of molecules, and to situations involving multiple reactions and multiple phases.

Work related to these chemical reaction simulation studies is the Gibbs ensemble technique of Panagiotopoulos,¹⁴ which was developed to simulate phase equilibria. As remarked above, phase equilibrium can be regarded as a trivial case of chemical reaction, according to



where S denotes a particular substance and α and β denote different phases. The Gibbs ensemble method can thereby

be regarded as a special case of our reaction ensemble approach, in which the only reactions in the system are those of the form of Eq. (6).

In Sec. II of this paper, we describe the basis of the reaction ensemble method. We then apply the method to several model single-phase reacting systems at either fixed (T, P) or (T, V) and compare our results to those obtained using accurate semiempirical equation-of-state methods, and to those of Shaw for reaction (5). As demonstrations of the relative simplicity of the method, the examples include systems involving more than one simultaneous reaction, and cases in which the total number of molecules is not conserved. The following sections discuss the results of our calculations, and their comparison with those obtained using semiempirical formulas and with those of other workers. In future papers, we intend to demonstrate the use of the approach for more complex systems, including additional model systems and systems of industrial interest, and those exhibiting combined phase and chemical reaction equilibria.

II. THE REACTION ENSEMBLE METHOD

We first derive the method in the case of a single chemical reaction occurring in one phase at specified (T, V) . We then generalize to allow for specified (T, P) and/or multiple reactions and phases. In Appendix A, we provide a formal derivation using a Legendre transformation approach.

We consider a system of s different chemical species, and begin by writing the usual canonical ensemble partition function for a *nonreacting* system. Within the approximation of separable molecular internal degrees of freedom, this is given by¹⁵

$$\begin{aligned} Q(N_1, N_2, \dots, N_s, V, T) \\ = \prod_{i=1}^s \frac{(Vq_i/\Lambda_i^3)^{N_i}}{N_i!} \\ \times \int \exp[-\beta U(V, \mathbf{z}_1, \dots, \mathbf{z}_N)] d\mathbf{z}_1, \dots, d\mathbf{z}_N, \end{aligned} \quad (7)$$

where V is the system volume, q_i is the part of the partition function corresponding to an isolated molecule, Λ_i is the de Broglie thermal wavelength, N_i is the number of molecules of species i , and $U(V, \mathbf{z}_1, \dots, \mathbf{z}_N)$ is the configurational energy expressed in terms of V and the relative coordinates $\mathbf{z}_i$ of the molecules, the components of which lie between 0 and 1. The total number of molecules in the system is

$$N = \sum_{i=1}^s N_i. \quad (8)$$

In the partition function (7), N_i is fixed.

The goal is to devise a scheme to express the analog of the partition function (7) in the case when the set of system species undergoes chemical reaction. Formally, this requires the summation of Eq. (7) over all sets of N_i subject to the constraint imposed by the reaction. The problem is to account for this constraint in a computationally efficient manner, which we will accomplish in the following by

means of a simple transformation of variables, based on the link between the stoichiometry and the elemental abundance constraints of Eq. (2).

Consider a general reacting system in which the molecules are comprised of m different atomic elements.^{16,17} The quantities a_{ji} ; $j=1,2,\dots,m$ specify the molecular composition of each species i , and the vector $\mathbf{a}_i$ is called the *formula vector* of species i . The collection of system formula vectors define the system *formula matrix* $\mathbf{A}$, given by

$$\mathbf{A} = (\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_s). \quad (9)$$

We consider here the general chemical reaction

$$\sum_{i=1}^s \nu_i \mathbf{a}_i = 0, \quad (10)$$

where $\{\nu_i\}$ are a set of stoichiometric coefficients and $\mathbf{a}_i$ is the vector of coefficients of the chemical formula of species i . For a given reaction, we adopt the usual convention that species for which $\nu_i < 0$ are called reactants and those for which $\nu_i > 0$ are called products. The fundamental property of $\{\nu_i\}$ is the fact that, via the orthogonality relation of Eq. (10), they embody the law of conservation of mass for the system, Eq. (2), which may be expressed on a molecular basis as

$$\sum_{i=1}^s a_{ji} N_i = B_j; \quad j=1,2,\dots,m, \quad (11)$$

where B_j is the total number of atoms of type j in the system. The set of stoichiometric coefficients obeying Eq. (10) allows all solutions of Eq. (11) to be expressible as

$$N_i = N_i^0 + \nu_i \xi, \quad (12)$$

where N_i^0 represents an arbitrary particular system composition satisfying Eq. (11), $\{\nu_i\}$ satisfy Eq. (10), and ξ is a scalar parameter called the *extent of reaction*. Equation (12) is the transformation that takes into account the reaction, as may be seen by substituting Eq. (12) into Eq. (11), which yields

$$\sum_{i=1}^s a_{ji} (N_i^0 + \nu_i \xi) = \sum_{i=1}^s a_{ji} N_i^0 + \xi \left(\sum_{i=1}^s \nu_i a_{ji} \right); \quad j=1,2,\dots,m. \quad (13)$$

The first term on the right-hand side is B_j , and the second term vanishes due to Eq. (10). We remark that, in a macroscopic system, ξ is a continuous variable, whereas at a molecular level it must take on only integer values.

For a reacting system, the partition function (7) must be evaluated by summing over all possible sets of N_i that satisfy the conservation of atomic elements for the system. This may be performed efficiently by transforming the variables N_i to the reaction coordinate ξ using Eq. (12). The partition function for the reacting system is then obtained by substituting Eq. (12) into Eq. (7) and summing over all (integral) values of ξ that give $n_i > 0$, giving

$$Q(\chi, V, T) = \sum_{\xi} \prod_{i=1}^s \frac{(V q_i / \Lambda_i^3)^{(N_i^0 + \xi \nu_i)}}{(N_i^0 + \xi \nu_i)!} \times \int \exp[-\beta U(V, \mathbf{z}_1, \dots, \mathbf{z}_N)] d\mathbf{z}_1, \dots, d\mathbf{z}_N, \quad (14)$$

where χ denotes the N_i^0 of the reacting system satisfying Eq. (11) and $\mathbf{z}$ are scaled variables in (0,1). The summation is over all (integral) values of ξ that permit the molecular amounts N_i to remain non-negative. Equation (14) is the desired partition function for a system at specified (V, T) undergoing a single chemical reaction.

The Markov chain required to simulate Eq. (14) consists of two types of state transitions. One consists of particle moves sampling the configurational space for a fixed system composition (the usual (N, V, T) sampling procedure), and the other consists of "reaction moves," sampling the ξ variable.

The latter sampling procedure must ultimately lead to the satisfaction of the reaction equilibrium conditions

$$\sum_{i=1}^s \nu_i \mu_i = 0. \quad (15)$$

We may avoid the calculation of chemical potentials themselves by instead considering "reaction steps," consisting of the simultaneous insertion and deletion of all the particles in the system, according to the stoichiometry of the reaction. The precise method of implementing a reaction step must be carefully considered, both to preserve the microscopic reversibility condition, and to endow the simulation with reasonable convergence properties. The primary task is to determine the acceptance probability for such a reaction move.

We consider transitions between states k and l in which the reaction proceeds either "forwards" ($\xi > 0$) or "backwards" ($\xi < 0$) by $|\xi|$ molecular units from a given initial state. In state k prior to the reaction step, the composition is denoted by N^0 and we set $\xi = 0$. The final compositional state l is given by Eq. (12).

In the following, we allow a reaction to potentially proceed in either the forward or reverse direction with equal probability 0.5. The implementation of the reaction step must be consistent with microscopic reversibility. Suppose, e.g., we wish to test if the reaction may proceed in a forward direction from a given state k . We first randomly select a particular set of "reactant" molecules (those with $\nu_i < 0$) from the set of all such reactant molecules in the simulation. State l arises from this state by replacing all molecules in the selected set by a set of "product" molecules (those with $\nu_i > 0$). This must be done in a fixed manner in the course of the simulation, in order to preserve the microscopic reversibility condition. If $\bar{v} = 0$, then a natural way to do this is to attempt to place the product molecules at the positions of the potentially removed reactant molecules. Convergence of the simulation is enhanced if molecules are replaced by those of similar sizes. In any

event, the forward and reverse reaction steps must be attempted in exactly the same way each time. For example, for the reaction



when we consider a forward reaction step, we first randomly select a pair of A and B molecules. In the potential final state, either A is replaced by C and B is replaced by D, or A is replaced by D and B is replaced by C. The particular choice will affect the convergence of the algorithm, but such a choice must be made at the outset of the simulation. Suppose, e.g., we have made the first choice. Then, when we attempt a backwards reaction step, we must always attempt to replace C by A and D by B.

If $\bar{v}$, the change in the total number of molecules per unit of reaction,

$$\bar{v} = \sum_{i=1}^s \nu_i, \quad (17)$$

is nonzero, then there will be a deficit or surplus of molecules on either the forward or reverse reaction steps. Suppose, e.g., that $\bar{v} > 0$, and we are considering a potential forward reaction step. Any additional molecules are inserted into the box at random positions. If $\bar{v} < 0$, some of the formerly occupied positions remain vacant. The best choice of particle correspondence is one when the corresponding particles are of similar size and the randomly inserted particles are as small as possible.

Given the above simulation scheme, we show in Appendix B that the required transition probability $k \rightarrow l$ is obtained directly from Eq. (14) as the ratio of terms in the partition function:

$$P_{kl}^{\xi} = \min \left[1, \prod_{i=1}^s \left[(V q_i / \Lambda_i^3)^{\nu_i^{\xi}} \frac{(N_i^0)!}{(N_i^0 + \nu_i^{\xi})!} \right] \exp(-\beta \Delta U_{kl}) \right] \\ = \min \left[1, V^{\bar{v}^{\xi}} \Gamma^{\xi} \prod_{i=1}^s \left[\frac{(N_i^0)!}{(N_i^0 + \nu_i^{\xi})!} \right] \exp(-\beta \Delta U_{kl}) \right], \quad (18)$$

where Γ is the ideal-gas quantity

$$\Gamma(T) = \prod_i \left(\frac{q_i}{\Lambda_i^3} \right)^{\nu_i} \quad (19)$$

and

$$\Delta U_{kl} = U_l - U_k. \quad (20)$$

Γ is related to tabular thermodynamic data¹⁸ by

$$\Gamma(T) = e^{-\Delta G^0(T)/RT} \left(\frac{P^0}{RT} \right)^{\bar{v}}, \quad (21)$$

where

$$\Delta G^0(T) = \sum_i \nu_i \Delta G_{fi}^0(T) \quad (22)$$

and $\Delta G_{fi}^0(T)$ is the ideal-gas standard free energy of formation of species i at the reference pressure P^0 . We remark that cases in which $(\bar{v} \neq 0)$ are simply and explicitly accounted for by the term in Eq. (18) involving V .

To simulate a chemical reaction at specified (P, T) , rather than at (T, V) , we must incorporate a third type of state transition in addition to the particle position and reaction moves, that of trial volume changes, $V_k \rightarrow V_l$. These trial moves are implemented in the usual way, and are governed by by¹⁹

$$\ln(P_{kl}^V) = \min \left[0, -\beta P(V_l - V_k) - \beta \Delta U_{kl} - N \ln \left(\frac{V_k}{V_l} \right) \right]. \quad (23)$$

The Gibbs ensemble method can be obtained as a special case of the reaction ensemble method by considering a two-phase system undergoing phase equilibrium reactions of the type given by Eq. (6), which can be written as

$$1 \cdot S_i - 1 \cdot S_j = 0, \quad (24)$$

where species i and j are in different phases and $\nu_i = 1$, $\nu_j = -1$. Each phase is modeled by a separate simulation box, and the reaction steps take place from one box to another. Thus, for phase equilibrium, $\Gamma(T) = 1$, $\bar{v} = 0$, and Eq. (18) becomes equivalent to the "particle transfer acceptance probability" in the Gibbs ensemble method. In addition to reaction moves to ensure ultimate equality of chemical potentials, the Gibbs ensemble method ensures equality of pressures in each phase by means of volume change steps for each simulation box, using the appropriate analog of Eq. (23) for the set of simulation boxes. This procedure must also be carried out when the reaction ensemble method is employed for multiphase systems.

Finally, we generalize our procedure to any linearly independent set of R chemical reactions, given by

$$\sum_{i=1}^s \nu_{ji} S_i = 0; \quad j = 1, 2, \dots, R. \quad (25)$$

The coordinate transformation from $\{N_i\}$ to $\{\xi_j\}$ is given by

$$N_i = N_i^0 + \sum_{j=1}^R \nu_{ji} \xi_j, \quad i = 1, 2, \dots, s; \quad j = 1, 2, \dots, R. \quad (26)$$

The resulting reaction partition function that satisfies the equilibrium conditions

$$\sum_{i=1}^s \nu_{ji} \mu_i = 0, \quad j = 1, 2, \dots, R \quad (27)$$

is

$$Q(\chi, V, T) \\ = \sum_{\xi_1} \sum_{\xi_2} \dots \sum_{\xi_R} \prod_{i=1}^s \frac{(V q_i / \Lambda_i^3)^{(N_i^0 + \sum_j \nu_{ji} \xi_j)}}{(N_i^0 + \sum_j \nu_{ji} \xi_j)!} \\ \times \int \exp[-\beta U(V, \mathbf{z}_1, \dots, \mathbf{z}_N)] d\mathbf{z}_1, \dots, d\mathbf{z}_N \quad (28)$$

and the transition probability for an individual reaction step ξ_j is

$$P_{kl}^{\xi_j} = \min \left\{ 1, V^{\bar{v}_{\xi_j}} \Gamma_j^{\xi_j} \prod_{i=1}^s \left[\frac{(N_i^0 + v_{ji} \xi_j)!}{(N_i^0)!} \right] \exp(-\beta \Delta U_{kl}) \right\}. \quad (29)$$

As in the case of systems involving only a single reaction, Eqs. (28) and (29) may be applied to the case of specified (P, T) by the incorporation of volume change moves, and to multiple phases by means of additional simulation boxes and the imposition of the constraint of equal pressures on each such box.

To summarize, the reaction ensemble method (for a single-phase system) consists of a combination of the following types of steps:

1. Particle displacement (NVT) steps.
2. Reaction steps, consisting of the following:
 - (a) Randomly select a reaction (if there are more than 1).
 - (b) Randomly select whether the reaction step will be forwards or backwards.
 - (c) Randomly select a set of reactant and product molecules in the system.
 - (d) Implement the reaction (χ, V, T) step according to Eqs. (29) and (12).
3. In the case of constant pressure simulation, attempt a volume change (NPT) step.

The convergence of the method is affected by the relative ratios of the three different types of steps, and the optimal choice depends on the system properties. The relative frequency of steps 1 and 3 should be chosen similarly to that used in conventional NPT simulations (i.e., $\approx N:1$, where N is the total number of particles in the system). We have found that it is reasonable to attempt, for each volume step, a sufficient number of reaction steps so that the number of accepted steps is several percent of the total number of molecules.

III. EXAMPLE SYSTEMS AND COMPUTATIONAL DETAILS

To illustrate the reaction ensemble method, we consider five different reacting systems. We include examples of cases considered difficult by previous workers to demonstrate the simplicity and flexibility of the method. Thus, two examples illustrate the case $\bar{v} \neq 0$, and one considers the case of two simultaneous reactions. The first four examples involve model hard-sphere systems. Although such systems are only tenuously related to real systems, they provide both an illustration and a verification of the method, since the equilibrium compositions may also be calculated independently using accurate semiempirical results for the hard-sphere systems. In fact, our results may be more accurately viewed as providing tests of these semiempirical equations. The final example is the NO formation reaction (5) considered previously by Shaw.¹¹⁻¹³

The first example system involves the isomerization reaction



where A and B are hard spheres of diameters σ_A and σ_B . For this system, we considered the case $\sigma_B/\sigma_A = 0.5$. For illustrative purposes, we set $\Gamma = 1$, and performed calculations as a function of overall density $\rho^* = N\sigma_A^3/V$, considering reaction at either constant volume or constant pressure.

The second example is the hard-sphere pseudodimerization system



We set

$$\frac{\sigma_B}{\sigma_A} = 2^{1/3} \quad (32)$$

and calculated the equilibrium composition as a function of density $\rho^* = \Sigma N_i \sigma_i^3/V$, for the case $\Gamma^* = \Gamma/\sigma_A^3 = 1$, and as a function of Γ^* for $\rho^* = 0.5$.

The third example is



where we chose

$$\frac{\sigma_B}{\sigma_A} = 2^{1/3}, \quad (34)$$

$$\frac{\sigma_C}{\sigma_A} = 3^{1/3} \quad (35)$$

and calculated the equilibrium composition as a function of overall reduced density for $\Gamma^* = 1$, for an overall composition corresponding to initially equimolar amounts of A and B.

The fourth example involves a system simultaneously undergoing both reactions (31) and (33) with $\Gamma^* = 1$ for both reactions, and the final example is the NO formation reaction (5) considered previously by Shaw.¹¹⁻¹³

We performed all our simulations employing the usual periodic boundary conditions, with spherical cutoffs in the potential for the case of (5). Based on previous experience with similar systems, in each simulation the total numbers of particles were chosen so that the average number of particles of any individual species was at least 30. The calculations were generally carried out so that approximately 30% of the simulation steps were reaction steps, 1% were volume change steps (for the constant P runs), and the remainder were particle translation steps. At high densities, the number of reaction steps was increased up to near 85%. All calculations were programmed in FORTRAN and run on a HP 9000/730 computer.

IV. RESULTS AND DISCUSSION

For the first four examples, involving hard-sphere systems, we compared our results with those obtained assuming an ideal solution and with results obtained using an equation of state for hard-sphere mixtures due to Boublik.²⁰

Our results for the isomerization system are presented in Fig. 1, where the equilibrium mole fraction of the larger molecule A is shown as a function of overall density. The results of calculations at constant overall density and con-

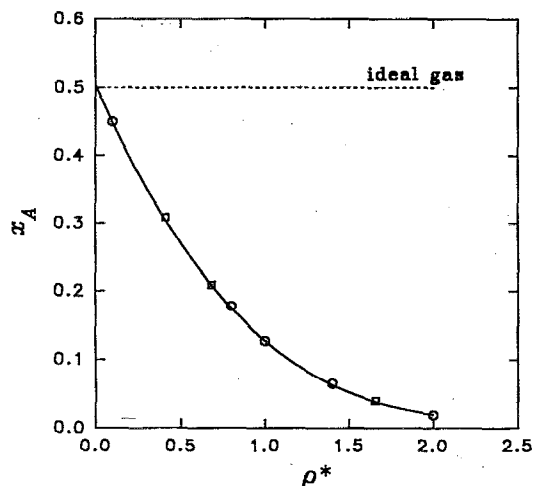


FIG. 1. Equilibrium for the hard-sphere isomerization reaction (30), as a function of overall density, with $\Gamma^*=1$. x_A is the equilibrium mole fraction of hard spheres with diameter σ_A . Circles denote calculations at constant (VT) and squares denote calculations at constant (PT). The solid curve results from the use of Boublik's (Ref. 20) equation of state and the dashed curve is the ideal-gas result.

stant pressure are both shown. It is seen that the results obtained numerically using Boublik's²⁰ equation of state agree excellently with those of the simulations, and that the ideal-gas results are considerably in error.

Our results for the pseudodimerization system are presented in Fig. 2. In Fig. 2(a), the equilibrium mole fraction of the monomer is shown as a function of overall density, for calculations at both constant overall density and constant pressure. In Fig. 2(b), the dependence on Γ^* is shown at a constant overall density of $\rho^*=0.5$. Again, the

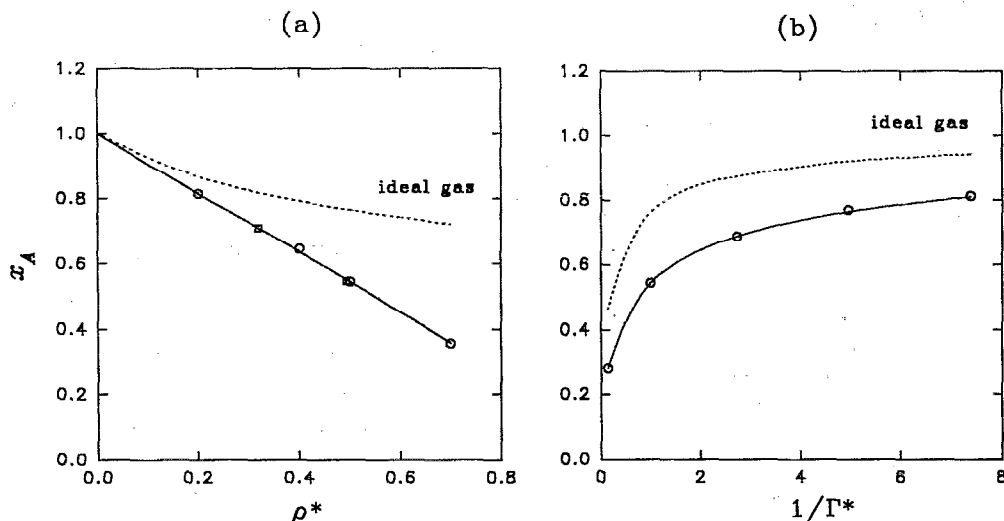


FIG. 2. Equilibrium for the hard-sphere pseudodimerization reaction (31). (a) Density dependence for $\Gamma^*=1$, and (b) dependence on Γ^* at fixed overall density. x_A is the equilibrium mole fraction of hard spheres with diameter σ_A . Circles denote calculations at constant (VT) and squares denote calculations at constant (PT). The solid curves result from the use of Boublik's (Ref. 20) equation of state and the dashed curves are the ideal-gas results.

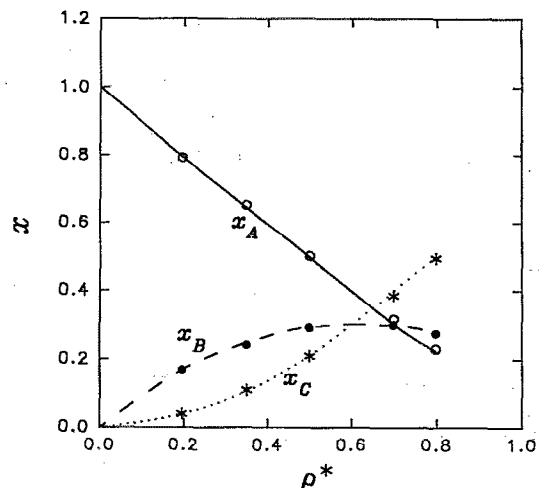


FIG. 3. Equilibrium for the hard-sphere reaction (33) as a function of overall density, for $\Gamma^*=1$, corresponding to initially equimolar amounts of A and B. x denotes the mole fractions of the three species and the points denote our simulation results. The curves result from the use of Boublik's (Ref. 20) equation of state.

equation of state results are excellent. The ideal-gas results are erroneous, but not as poor as in the case of the isomerization system.

Our results for the reaction of Eq. (33) are presented in Fig. 3, where we show the three equilibrium mole fractions as a function of both reduced density and Γ^* corresponding to an equimolar reactant overall composition. The results obtained from the equation of state are excellent everywhere, and only slightly in error at the highest density.

Our results for the two-reaction system are presented

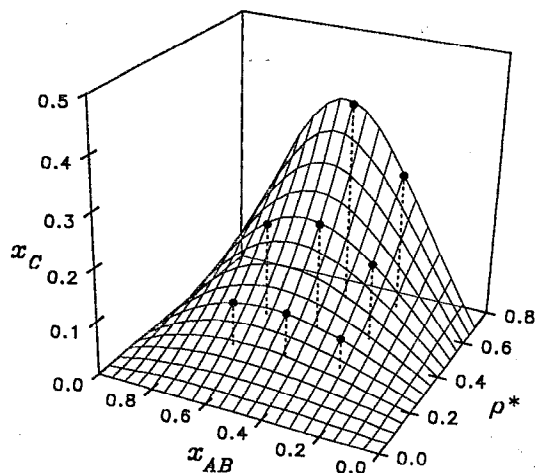


FIG. 4. Equilibrium for the hard-sphere two-reaction system, reactions (31) and (33), for $\Gamma^*=1$, as a function of overall density and x_{AB} , the overall ratio of amounts of species A and B. x_C is the mole fraction of species C, and the points denote our simulation results. The curves result from the use of Boublik's (Ref. 20) equation of state.

in Fig. 4, where we show the equilibrium mole fractions as a function of reduced density for the case of $\Gamma^*=1$ for both reactions. The equation of state results are again excellent.

Our results for the system simulated by Shaw¹¹⁻¹³ are given in Table I. It is seen that the results at $x_O=0.5$ are in essentially exact agreement. Our more accurate results using larger numbers of molecules than used by Shaw indicate his results for the equilibrium NO mole fractions are slightly high.

V. CONCLUSIONS

We have derived a new Monte Carlo simulation method called the reaction ensemble method, that is computationally well-suited for the simulation of equilibria in systems involving any number of chemical reactions existing in any number of different phases. This method allows the efficient determination of chemical reaction equilibrium, including combined reaction and phase equilibrium, directly from a description of the intermolecular interactions among the molecules of the system.

TABLE I. Simulation of reaction (5).

x_O^a	N^b	This paper	Shaw ^c
1/6	120		0.1789(08)
1/6	402	0.1785(04)	
1/3	120		0.2549(16)
1/3	399	0.2522(05)	
1/2	120	0.2763(04)	0.2768(22)
1/2	400	0.2755(04)	
2/3	120		0.2555(17)
2/3	399	0.2543(04)	
5/6	120		0.1828(11)
5/6	402	0.1810(04)	

^aOverall mole fraction of O.

^bTotal number of particles in the simulation.

^cReference 11. Numbers in parentheses denote precisions.

The reaction ensemble method is an improvement on the approach of Kofke and Glandt,¹⁰ due to the fact that chemical potentials and/or fugacities need not be calculated. Previously the most important available method for simulating chemical reactions is that due to Shaw.¹¹⁻¹³ Our method differs from his approach, both in its microscopic viewpoint of particle distinguishability and also in its exploitation of the intrinsic link between the chemical stoichiometry and the underlying constraints of conservation of mass. This results in an extremely simple set of computational formulas, which are well-suited to many cases of reaction equilibria considered too complex for straightforward treatment in the past. This includes the cases of multiple reactions and reactions for which the total number of molecules is not conserved, as well as reactions in multi-phase systems.

We have derived the reaction ensemble method here for reacting systems at either fixed (T,V) or fixed (T,P) , although it may be easily extended to cases of other sets of thermodynamic constraints. The Gibbs ensemble method may be viewed as a special case of the reaction ensemble method, in which the only "reactions" take place between species in different phases.

We have illustrated the reaction ensemble method for four model systems of reacting hard spheres, for which results may also be obtained using alternative approaches based on semiempirical equations of state. Our calculations may be viewed as tests of the accuracy of these semiempirical equations. We have also considered a more realistic system, modeling the formation of NO from O₂ and N₂, an example that has been considered previously.¹¹⁻¹³

In future papers, we plan to use the method to perform calculations for more complex model and realistic systems involving combined phase and reaction equilibria.

APPENDIX A: REACTION ENSEMBLE PARTITION FUNCTION FROM A LEGENDRE TRANSFORMATION

We consider a system undergoing the single chemical reaction of Eq. (10), and generalize our results to the case of several simultaneous reactions. Chemical equilibrium occurs when

$$\sum_{i=1}^s \nu_i \mu_i = 0. \quad (\text{A1})$$

This condition can be considered as an external constraint imposed on our thermodynamic system and our task is to employ this constraint. Our derivation will show that the fact that the right-hand side of Eq. (A1) is exactly zero is unimportant. Theoretically, it can be held fixed at any constant value. (This situation has important applications to some systems of chemical interest.²¹)

The thermodynamics of a system at constant T,V and total numbers of all types of molecules N_i can be described

by a canonical ensemble whose corresponding thermodynamic potential is the Helmholtz free energy A , with the corresponding fundamental equation

$$dA = -S dT - P dV + \sum_{i=1}^s \mu_i dN_i, \quad (\text{A2})$$

where S is the entropy. The partition function Q of the canonical ensemble is given by

$$Q = \prod_{i=1}^s \frac{(Vq_i/\Lambda_i)^{N_i}}{N_i!} \times \int \exp[-\beta U(V, z_1, \dots, z_N)] dz_1, \dots, dz_N, \quad (\text{A3})$$

where U is the total potential energy, z_i are relative coordinates of the particles in the simulation box, q_i are the internal contributions to the partition function, Λ_i are de Broglie thermal wavelengths, and $N = \sum N_i$.

This ensemble is not convenient for the simulation of chemical equilibria because the numbers of all the molecules which take part in the chemical reaction (have non-zero v_i) are variable, whereas the intensive variable $\sum v_i \mu_i$ is constant. We perform a Legendre transformation to create a new thermodynamic potential whose natural variables include $\sum v_i \mu_i$. We first perform a transformation of coordinates in the space of the variables N_i , $i = 1, \dots, s$. The most suitable form of such a transformation is

$$N_i = v_i \xi + Y_i(b_1, b_2, \dots, b_{s-1}), \quad (\text{A4})$$

where ξ is the extent of the chemical reaction and Y_i is a function of $(s-1)$ auxiliary parameters that depend on the overall system composition. In the following, the exact definition of the quantities Y_i is irrelevant, since they will ultimately be held fixed due to the imposition of the closed system constraint, and they do not need to be subjected to the Legendre transformation. The new form of the fundamental equation, which is exactly equivalent to Eq. (A2) is

$$dA = -S dT - P dV + \sum_{i=1}^s v_i \mu_i d\xi + \sum_{i=1}^s \sum_{j=1}^{s-1} \mu_i \frac{\partial Y_i}{\partial b_j} db_j. \quad (\text{A5})$$

Now we may define a new thermodynamic potential A^X by the equation

$$A^X = A - \xi \sum_{i=1}^s v_i \mu_i \quad (\text{A6})$$

whose infinitesimal change can be written as

$$dA^X = -S dT - P dV - \xi d\left(\sum_{i=1}^s v_i \mu_i\right) + \sum_{i=1}^s \sum_{j=1}^{s-1} \mu_i \frac{\partial Y_i}{\partial b_j} db_j. \quad (\text{A7})$$

The variable $\sum v_i \mu_i$ plays the role of a new natural variable of A^X , so it can be held fixed during the simulation. The partition function corresponding to the potential A^X is

given by the usual summation¹⁵ of the canonical partition functions (A3) over all possible values of the extensive variable ξ :

$$Q^X = \sum_{\xi=1}^s \prod_{i=1}^s \frac{(Vq_i/\Lambda_i)^{N_i(\xi)}}{N_i(\xi)!} \times \int \exp[-\beta U(z_1, \dots, z_{N(\xi)})] dz_1, \dots, dz_{N(\xi)} \times \exp\left[\beta \xi \sum_{j=1}^s v_j \mu_j\right]. \quad (\text{A8})$$

If the system is at chemical equilibrium [satisfies Eq. (A1)], then the final exponential term in the partition function is equal to unity, in which case the potential A^X is numerically equal to A . We note, however, that we may, in principal, perform simulations on systems for which the summation is an arbitrary constant value.

This *reaction ensemble* is suitable for simulations when the natural coordinates of the thermodynamic potential A^X are held constant, which are T and V . The generalization to the case of constant pressure can be done in the usual way. A Legendre transformation is made by adding PV to A^X and integrating the reaction partition function over all possible values of the extensive variable V when multiplying each term by $\exp[-\beta PV]$.

In the case of more than one chemical reaction, the only difference is that the number of coordinates explicitly defined as *extents of reactions* in Eq. (A4) must be identical to the number of reactions, and all these extensive coordinates must be subjected to the Legendre transformation. The simulation steps will then include changes of all *extents of reaction* randomly selected.

APPENDIX B: PROOF OF MICROSCOPIC REVERSIBILITY FOR THE REACTION ENSEMBLE SIMULATION

The purpose of this Appendix is to prove that the proper ratio of forwards and backwards acceptance probabilities for a trial reaction step from one state to another is equal to the ratio of terms that correspond to both states in the partition function for a molecular system represented by a system of distinguishable particles. The case is quite similar to that of the grand canonical ensemble.

We shall suppose that in the simulation algorithm the probability that the trial step forwards is 1/2 and the probability that the reaction step backwards is also 1/2. The *a priori* probability that from the present state r the trial reaction step will lead to the state s is denoted by $p_{r \rightarrow s}^{(a)}$. The term corresponding to a given state r in the partition function is the term that, after dividing by the partition function Q , becomes the probability density. We shall denote this term by ρ_r , and the probability of this state $W(r)$ is

$$W(r) = \frac{\rho_r \delta^N}{Q}, \quad (\text{B1})$$

where δ is the minimum distinguishable volume element in our unit simulation cube containing the variables z_i (given

by the number of valid digits in our computer) and N is the total number of particles. In the isobaric ensemble an additional δ_V is necessary due to the integration over the extensive variable of volume. The ratio of forwards and backwards acceptance probabilities $w_{rs} = P_{rs}^{\xi} / P_{sr}^{-\xi}$ of a step in the Markov chain must be given by

$$w_{rs} = \frac{W(s) p_{s \rightarrow r}^{(a)}}{W(r) p_{r \rightarrow s}^{(a)}} \quad (\text{B2})$$

We shall assume in the following that species with indices $i=1, \dots, l$ are on the left-hand side of the equation (reactants) and species with indices $i=l+1, \dots, s$ are on the right-hand side (products). We shall denote by N_i the numbers of molecules of each species in state r and assume that the reaction step is from the left to the right with the number of product particles greater than or equal to the number of reactants. Other cases are treated similarly.

If we have decided that the reaction step will take place from left to right, we must randomly select $|v_i|$ reactant

molecules (those for which $v_i < 0$) of each type $i=1, \dots, l$, and change their identities according to some scheme that we have adopted, to product molecules of types $i=l+1, \dots, s$. If $\bar{v} < 0$, the remaining particles are inserted into the simulation box at random positions. In the opposite case, some of the places previously occupied by reactant molecules remain vacant. We have $N_1(N_1-1) \cdots (N_1 + v_1 + 1)$ possible ways to select particle number 1 and analogous numbers of ways for other reactant particles. If we select them and change them to product particles according to our scheme, the first particle of type s can be assigned any label between 1 and $N_s + 1$ (because the distinguishable particles form an ordered set). Similarly, the number of possible positions can be calculated for all particles $i=l+1, \dots, s-1$. If some particles are added at random, the probability that they will fall exactly in a definite volume element is δ , since the volume of the unit cube is unity. The *a priori* probability that a particular state s will be selected as the result of a reaction step from state r is

$$p_{r \rightarrow s}^{(a)} = \frac{\delta^{\bar{v}}}{2N_1 \cdots (N_1 + v_1 + 1) \cdots N_l \cdots (N_l + v_l + 1) (N_{l+1} + 1) \cdots (N_{l+1} + v_{l+1} + 1) \cdots (N_s + 1) \cdots (N_s + v_s)} \quad (\text{B3})$$

The *a priori* probability of a reverse step from state s to state r differs only in two aspects. First, particles with labels $i=1, \dots, l$ are created and those with labels $i=l+1, \dots, s$ are destroyed. Second, the numbers of particles of each type are increased by v_i from state r . The reverse *a priori* probability is therefore

$$p_{s \rightarrow r}^{(a)} = \frac{1}{2N_1 \cdots (N_1 + v_1 + 1) \cdots N_l \cdots (N_l + v_l + 1) (N_{l+1} + 1) \cdots (N_{l+1} + v_{l+1} + 1) \cdots (N_s + 1) \cdots (N_s + v_s)} \quad (\text{B4})$$

Substituting Eqs. (B1), (B3), and (B4) into Eq. (B2) we get for w_{rs} the result

$$w_{rs} = \delta^{-\bar{v}} \frac{W(s)}{W(r)} = \frac{\rho_s}{\rho_r} \quad (\text{B5})$$

To preserve microscopic reversibility, it is necessary to carefully consider the particle correspondence in the forward and backward reaction step. If in the forward step a particle B is placed at the position of a particle A, then in the backward step particle A must be placed at the former position of a particle B.

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